

# Cultivating self-reliance? An analysis of Uganda's refugee land allocation policy

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## Abstract

This paper examines the impact of Uganda's "self-reliance" policy for refugees, which includes allocating land to new refugee arrivals with the intention of reducing household reliance on international aid. Using household survey data covering 9 settlements from 2017 to 2021, along with statistical methods to circumvent bias in the estimation, we estimate the effect of land allocations upon arrival on household food security and agricultural production outcomes. We find that only 40 percent of refugee households actually receive a land allocation greater than or equal to 0.25 acres upon arrival at a settlement. Our estimations show that larger initial land allocations are associated with higher food security, greater propensity to harvest and to have any agricultural earnings, more land under cultivation, and a higher likelihood of having any livestock. However, we also demonstrate that in order to improve refugee outcomes so that they are similar to the surrounding host communities, refugee land allocation would have to rise from 0.25 acres to 1.84 acres. This level of land transfer would exceed the total area of land officially set aside in many refugee settlements by a substantial amount, even before taking into account variation in land quality. Qualitative interviews suggest that relatively low agricultural production among refugees may be due to the fact that many had never farmed until they arrived in Uganda, and that there is little access to agricultural inputs. We conclude that while the policy does improve refugee welfare, self-reliance through land allocation alone is insufficient without concurrent support in refugees' agricultural capacity.

**Significance statement:** Uganda's policy of giving land to refugees upon arrival in the country is considered a positive example of how to manage displaced populations. We examine agricultural and food security outcomes for refugees in Ugandan settlements between 2017 and 2021, and demonstrate that those who received more land adopted fewer hunger coping strategies and produced more food. However, their outcomes were substantially worse than Ugandan hosts living in the same areas, who are also very poor. For land allocation to allow refugees to reach equivalent food security as hosts, they would require 7.4 times more land than the stated 0.25-acre target. This would largely exceed the amount of land available for reallocation, demonstrating need for parallel investment in refugee livelihoods.

**Keywords:** Land allocation to refugees; Refugee policy; Uganda; Refugee self-reliance

# 1 Introduction

The global number of forcibly displaced people continues to increase at a sobering pace, more than doubling between 2012 and 2022 (1). The majority of these families have migrated to neighboring low-income countries where governments have few resources to dedicate to their support (2). Countries have taken divergent policy approaches towards refugee hosting, ranging from not recognizing them as populations worthy of support to allocating significant rights and resources (1). Among the latter, Uganda stands out as unusually generous in its official stance towards displaced populations, including allocating refugees land when it is available. The distribution of land is meant to support a broader strategy of reducing refugee reliance on external aid (3). While this is a laudable goal, the implementation of this policy may be challenged by a number of factors, including lack of available land and limited refugee familiarity with farming.

The purpose of this paper is to assess the effectiveness of the land allocation policy in supporting refugee welfare and livelihoods. We do this by examining food security and agricultural production outcomes from refugees and hosts living near refugee settlements in Uganda between 2017 and 2021. Using statistical methods to eliminate bias from the estimation, we find that while the policy of giving plots of land is not universally applied, it does improve the food security and increase agricultural production among those who receive land. However, we also demonstrate that the amount of land required to lift refugee food security to the level of neighboring hosts is significantly larger than the arable land that is available to distribute in many settlements.

Our work forms part of a broader literature assessing strategies intended to generate greater self-reliance among refugee populations. This includes a recent experiment in shifting from food to cash aid and encouraging refugees to cultivate home gardens in Kenya (4), which demonstrates that this combined strategy has improved refugee outcomes. We focus here specifically on the role of the land given to refugees. Although there is significant documentation of the importance of land as a productive asset in low-income settings (5, 6, 7), there is currently little evidence in the refugee settlement context. The one exception to this is Zhu et al. (8), who use a general equilibrium model alongside cross-sectional analysis of data from a single settlement in Uganda to

demonstrate that providing refugees with land improved their welfare and self-reliance. A report on self-reliance among Somali and Congolese refugees in Uganda compares their outcomes to those of refugees in Kenya and finds that the combination of aid, potential labor mobility, and land allocation results in greater mobility, higher incomes, and possibly superior employment outcomes than refugees in Kenya whose movements are more restricted (9). Descriptive analysis of refugees using the same dataset that we present here shows that household food security outcomes vary with the level of aid transfers, which interact with the amount of land that they farm (10). Our work differs from these foundational studies by causally estimating the effect of the initial government land allocation itself, using a strategy that directly addresses the factors that might generate bias in these estimates. Causal estimates are essential for assessing whether the existing policy should be scaled up or potentially applied in other contexts.

## 2 Study design

Uganda has long been the largest host of refugees in Africa (11). It neighbors two countries with the most prolonged and largest humanitarian crises on the continent – South Sudan and the Democratic Republic of Congo (DRC) (12). Together, these crises have forcibly displaced more than 20 million people. Uganda currently hosts nearly 2 million refugees among its total population of around 51 million people (13), and is widely regarded as a model host due to its unique policies. Refugees in Uganda move freely, can access social services, receive aid of various types, and upon arrival may receive plots of land for cultivation (3). When refugees arrive in Uganda, they are registered by the UNHCR at a border crossing checkpoint and then directed to a specific settlement. Assignment of a refugee to a specific settlement is based on perceived land availability, aid, settlement crowding and overall absorption capacity (8). Once they arrive at a settlement, they may receive a plot of land for food production, although this is not guaranteed (see Figures S1, S2 and S3). The system is structured such that food aid is available only for an initial period after a refugee household’s arrival in the country, after which time the intention is that they become “self-

reliant”, potentially by using their allocated plots as a main source of livelihood (14) along with a limited stipend of cash aid. Our intention is to test whether allocating land leads to manifestations of self-reliance.

We examine this question using data from refugee settlements across Uganda. These surveys are implemented jointly by the Office of the Prime Minister of Uganda (OPM) and the Food and Agriculture Organization (FAO), using an instrument developed by FAO to measure food security known as the Resilience Index Measurement and Analysis (RIMA) survey. The sample is representative at the district and settlement levels and comprises 0.89 percent of the refugee population in the country (15). The survey contains information on basic household demographic information, including the year in which they arrived, measures of agricultural production, food security, and accessibility to markets and other infrastructure. Figure 1 shows the location of the RIMA-surveyed settlements and highlights the settlements used in our analysis.

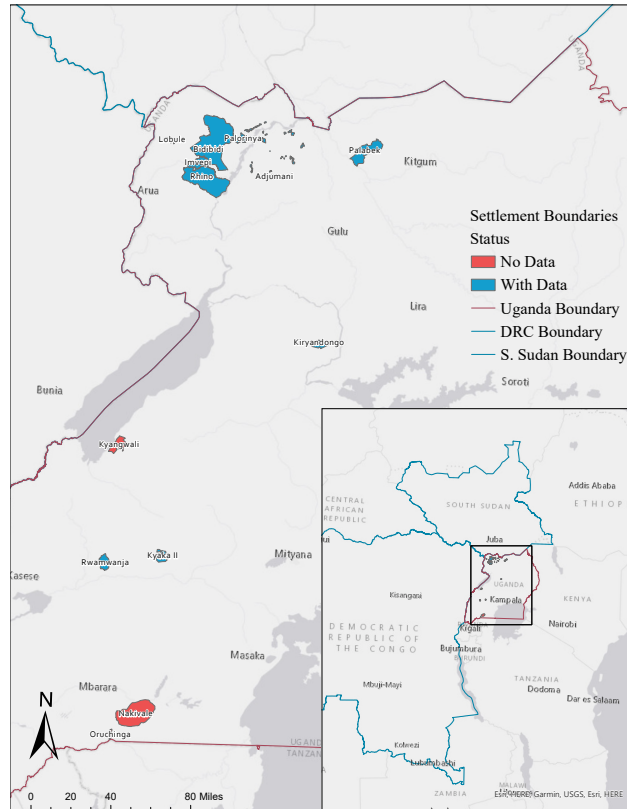


Figure 1: **Geographic distribution of refugee settlements in Uganda.** The main map displays the locations of the 12 refugee settlements. Settlements included in the primary analysis ( $N = 9$ ) are distinguished by color from those excluded due to data limitations ( $N = 3$ ). The inset map (corner) illustrates Uganda's location within East Africa.

Our analysis utilizes the subset of refugee households sampled in the first survey wave (2017-2019). We restricted the sample to the nine of twelve surveyed settlements that included specific questions regarding direct government land allocation, resulting in a final sample of 5,566 observations (reduced from 8,577). Households reported the size and acquisition method for each land parcel, allowing us to identify plots “received from the government for refugee status.” We use this question to proxy for the government policy of allocating land to refugees. We assume the area of these parcels reflects initial policy allocations, as refugees typically receive government land upon arrival; subsequent changes in tenure generally arise from renting or market transactions.

Table 1 shows characteristics of the analysis sample on average and stratified by initial government land allocation (above versus below 0.25 acres). This threshold is selected to distinguish households that align with the official settlement policy of 0.25 acres per household (3) from those with smaller plots. According to the 2019 Annual Agricultural Survey, the average landholding of an agricultural household in Uganda was 3.2 acres (16).

Table 1: Household Characteristics by Land Allocation

|                                   | Full sample | < 0.25 acres | $\geq$ 0.25 acres | Norm. Diff. |
|-----------------------------------|-------------|--------------|-------------------|-------------|
| <i>Land allocation</i>            |             |              |                   |             |
| Allocated $\geq$ 0.25 acres       | 0.40        |              |                   |             |
| Gov't Land Allocation (Acres)     | 0.38        | 0.12         | 0.77              | 1.61        |
| <i>Household characteristics</i>  |             |              |                   |             |
| Waves Surveyed                    | 3.19        | 3.14         | 3.26              | 0.14        |
| Arrival Year                      | 2014.77     | 2015.69      | 2011.31           | -0.67       |
| Household Size                    | 6.78        | 6.81         | 6.74              | -0.02       |
| Male HH Head                      | 0.56        | 0.57         | 0.54              | -0.06       |
| Years of School                   | 8.77        | 8.88         | 8.61              | -0.06       |
| Farming HH                        | 0.44        | 0.45         | 0.43              | -0.05       |
| Pastoralist HH                    | 0.15        | 0.11         | 0.22              | 0.31        |
| Cash Aid (USD), 12 mos            | 144.09      | 91.83        | 219.84            | 0.62        |
| In-Kind Aid (USD), 12 mos         | 143.96      | 176.48       | 86.81             | -0.47       |
| <i>Outcomes</i>                   |             |              |                   |             |
| Coping Strategies Index           | 14.25       | 14.38        | 14.04             | -0.03       |
| Total Crop Harvest (KG), 12 mos   | 209.21      | 140.84       | 313.03            | 0.44        |
| Total Crop Sales (KG), 12 mos     | 51.05       | 23.97        | 92.17             | 0.42        |
| Total Crop Earnings (USD), 12 mos | 12.51       | 7.26         | 20.49             | 0.31        |
| Crop Earnings Ind.                | 0.30        | 0.23         | 0.41              | 0.40        |
| Total Land Use (Acres)            | 0.69        | 0.60         | 0.83              | 0.21        |
| Normalized Asset Index            | 0.02        | -0.04        | 0.12              | 0.18        |
| Heads of Livestock Owned          | 1.88        | 1.88         | 1.88              | 0.00        |
| Livestock Earnings (USD), 12 mos  | 4.90        | 4.71         | 5.19              | 0.03        |
| N                                 | 5,566       | 3,365        | 2,201             |             |

*Notes:* The table includes mean values for the full analysis sample – column (1), the sample with land allocations less than 0.25 acres – column (2) and the sample with land allocations greater or equal to 0.25 acres – column (3). Column (4) reports the normalized difference in means between columns (2) and (3), which should be interpreted as the difference in standard deviations. The sample corresponds to the main analysis sample unless otherwise noted. “HH” corresponds to the household.



40% of the sample falls into the category of having been allocated more than 0.25 acres of land. This is the first finding of our paper: the land allocation policy has not universally benefitted refugees across Uganda. Among those allocated larger parcels of land, the mean allocated land (winsorized at the 99th percentile) is less than one acre. Among those with less than a quarter acre allocation, the mean is 0.12 acres.

Households average slightly less than seven members and just under nine years of schooling, with the majority reporting farming as their primary livelihood. The main observable driver of landholding variation is that households who have larger parcels of land tend to have arrived sooner (see Figures S1 and S2). These households with longer tenure also appear to be receiving more cash and less food aid than households with smaller parcels. The structure of the aid and the fact that larger parcel allocations are associated with more time in the country are consistent with the policy to make land allocations based upon land availability at the time of arrival – there was likely more land available to those who arrived sooner – and also that the mix of aid received is supposed to convert away from aid in kind and towards cash assistance with time. Interviews with settlement leadership corroborate this administrative rationing. As a UNHCR administrator that we interviewed in Nakivale settlement explained: “Originally, we gave 100 by 100-meter plots. This was later reduced to 50 by 50 meters, then 50 by 30 meters. Currently, new arrivals are allocated a 30 by 30-meter plot” (17).

Households with larger initial land allocations report fewer coping strategies (greater food security), harvest and sell more crops, and are more likely to report positive agricultural earnings as well as higher overall earnings. They also appear to farm more land – this number can be larger than the allocated parcel area because a parcel has been farmed multiple times in a year or the household acquired additional land through purchase or rental. Finally, we see no difference in total livestock owned, but those with more allocated land tend to have higher earnings and a larger asset index.

These differences should not be interpreted as causal effects of land allocation. A causal analysis requires understanding how the process of land assignment may bias simple correlations, and

finding a strategy to circumvent this bias. For example, the practice of assigning refugees to settlements based upon capacity and land availability may lead to refugees being assigned to locations that have better food security at the time of their arrival. The lack of randomness in the allocation process is not a problem if it is uncorrelated with a refugee’s ability to produce food on this land. However, if it is the case that land is systematically allocated to refugees with lower ability in agricultural production, then a simple regression analysis may show that receiving land reduces food security. Conversely, if land is systematically allocated to individuals who are highly skilled farmers, then we might observe a correlation that overstates the relationship between land allocation and food security.

To address this issue, we apply what is known as an “instrumental variables” estimation. This strategy uses a variable that affects the allocation of land within a settlement but does not directly affect a household’s ability to produce or procure food. The instrumental variable is used to generate a predicted measure of land allocation that is purged of factors that might confound the land allocation process with the household’s farming ability. This approach has a long history in the field of applied economics, the earliest known application being in 1928 (18). In our case, we use as an instrument the average land area allocated to refugees arriving in the same year (cohort) and settlement as the household, excluding the household’s own land allocation. This strategy relies on the assumption that, conditional on controls, cohort-level land allocations influence household outcomes only through the household’s own individual land allocation. Our controls include settlement, year and settlement-year fixed effects, which absorb both invariant and time-varying settlement shocks and fluctuations in aggregate production, average land allocations and local food prices. There is a potential concern regarding within-cohort spillovers, such as food sharing from higher allocation households to households perceived as food insecure. We discuss this possibility in the materials and methods section.

### 3 Results

Figure 2 displays the point estimates and 95% confidence intervals for key food security and production outcomes, comparing the instrumental variables (IV) and naive OLS approaches. We find that larger initial land allocations significantly improve food security.<sup>1</sup> Specifically, an allocation of 0.25 acres is associated with a food security increase of 0.07 standard deviations.

We next examine potential mechanisms driving this improvement. We find no effect on the likelihood of harvesting any crops. However, among households that do harvest, a 0.25-acre increase in land allocation is associated with an 18.2% increase in total harvest quantity (kg). Similarly, we observe increases in both the probability of reporting any earnings (5.1 pp per quarter acre) and the magnitude of agricultural earnings for households with positive income (21.3% per quarter acre).

The next three variables – total area of land used in the last 12 months, likelihood of owning livestock and the number of livestock – are meant to proxy for inputs into production and stores of household wealth. All of these are significantly higher for households that received larger land allocations from the government in the first wave of the survey. Despite these gains in production and inputs, we do not observe a significant increase in overall household wealth as measured by the asset index.

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<sup>1</sup>For ease of interpretation we reverse the Coping Strategies Index (CSI); higher values indicate greater food security.

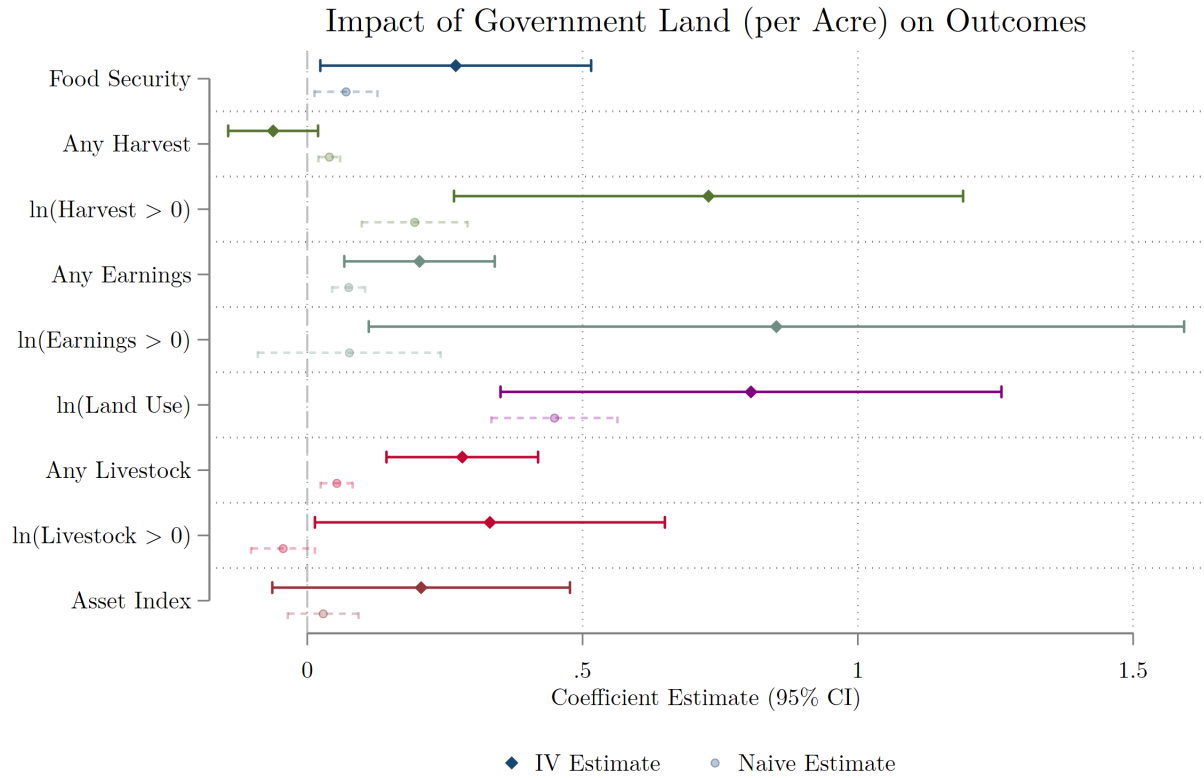


Figure 2: **Impact of Land Allocation on Household Outcomes.** The figure displays point estimates (dots) and 95% confidence intervals (lines) derived from the estimates in Tables S1 and S2. Results compare naive OLS and instrumental variable estimates, controlling for settlement-by-year fixed effects, weather measures, and household-level characteristics.

Comparison between the IV estimate and naive ordinary least squares (OLS) can give insight into the targeting of land allocations to refugees. The difference between these estimates is a combination of the bias that is caused by correlation between the types of individuals who receive land and the amount of land they receive, as well as the correlation between these types of individuals and the ability to produce or procure food. For example, if it is the case that larger plots of land are given to better farmers, then the naive, ordinary least squares regression results would be larger than those that have been corrected using IV. Comparison of these results with naive results that do not use an instrumental variable show that the IV estimates are almost always larger than the unadjusted regression point estimates (Figure 2, Table S1, and S2). This implies negative selection bias: land allocations tend to reach households with a relatively lower capacity to produce or procure food. This same relationship may also result from measurement error in the land allocation variable.

The explanation that land is allocated to farmers who may be less productive is consistent with interviews conducted by the research team in Uganda in September 2025, in which refugees cited a lack of agricultural experience and capital as major barriers to sufficient production. Refugees frequently reported limitations in both farming education and access to physical inputs. In Imvepi, respondents noted that “many of us were not farmers before,” and emphasized that “the methods we used 50 years ago are not enough anymore.” Furthermore, households stated that they are “lacking basic tools” like hoes and seeds, in addition to facing poor soil quality and a lack of irrigation systems (17). Many refugees have come from urban or non-farm backgrounds. Another from Imvepi settlement discussed their lack of experience: “I am a businessman. I am not used to digging. It is very hard for me”. A refugee in Bidi-Bidi stated, “Back home, I was a student. I have no interest in farming, but I have to do it to survive.” (17).

Given our finding that land is often allocated to households with lower initial farming capability, we examine whether the benefits of these allocations increase over time as refugees learn how to farm. To test this, we expand our main estimation (Eq. 2) by interacting the instrumented land allocation with a “Late Period” indicator (defined as post-2019). Table S3 reports the total

estimated effects of land allocation for the early and late periods in rows 1 and 3, respectively. We find that the impact on food security is immediate but attenuates moderately in the later period. Similarly, the effects on the intensive margin of earnings, likelihood of livestock ownership and total heads of livestock show a modest reduction in the late period compared to the early period. In contrast, the effects on total land use, likelihood of earnings and the intensive margin of harvest increase over time. These outcomes likely rely on the long-term accumulation of both physical and human capital to fully actualize the land's potential.

## 4 Discussion

The results demonstrate that Uganda's policy of giving land to refugees enhances the food security and ability to engage in agricultural markets of households that receive this land. They also show that the policy has reached less than half of the refugee population. This suggests that there is scope to improve refugee well-being by increasing the proportion of refugees who receive this land.

However, there may be political and practical constraints to doing this. First, allocating 0.25 acres to each of the approximately 265 thousand refugee households living in Uganda in 2020 would mean giving away 66,250 acres of land. The total land area of the refugee hosting districts in the country is around 5.8 million acres. This would mean transferring about 1.14 percent of the existing land area in these districts to refugees. Given rising tensions in Northern Uganda between refugee and host populations, this may be a difficult policy to implement. A settlement official in Imvepi noted the dwindling support by host communities, stating that "They (host communities) are reluctant to offer more land" (17).

To contextualize the magnitude of these constraints, we calculate the land allocation required for the average refugee household to achieve parity with their host-community neighbors. Table 2 presents the mean outcomes for both the refugee and host populations in the sample, along with the normalized difference of these means. The fourth column shows, based upon the marginal effects estimated in our main results, the size of the land allocation that would elevate the refugee

outcomes to the level of the hosts'. Across all statistically significant point estimates from the main results, the current mean allocation of 0.38 acres—and the highest settlement-level average of 0.97 acres (see Figure S3)—falls drastically short of the land required to match host outcomes. This gap is particularly striking given that host communities themselves already face high levels of poverty and food insecurity (15). We estimate that to achieve parity in food security, refugees would need an average allocation of 1.84 acres. For agricultural production (harvest volume, probability of earnings, and total earnings), parity requires allocations ranging from 1.05 to 1.58 acres. Similarly, equivalence in wealth and asset indicators, such as total land use and livestock ownership, would require between 0.80 and 1.60 acres.

Table 2: Comparison of Average Refugee and Host Outcomes

|                             | (1)<br>Refugee | (2)<br>Host | (3)<br>Norm. Diff. | (4)<br>Parity (Acres) |
|-----------------------------|----------------|-------------|--------------------|-----------------------|
| Gov't Land Area (Acres)     | 0.38           |             |                    |                       |
| Coping Strategy Index       | 14.25          | 8.84        | 0.54               | 1.84                  |
| Any Harvest (Indicator)     | 0.86           | 0.96        | -0.36              | .                     |
| Harvest (KG)                | 209.21         | 742.51      | -0.87              | 1.58                  |
| Any Ag Earnings (Indicator) | 0.30           | 0.61        | -0.65              | 1.51                  |
| Ag Earnings (USD)           | 12.51          | 67.12       | -0.72              | 1.05                  |
| Total Land Use (Acres)      | 0.69           | 2.77        | -1.24              | 1.60                  |
| Any Livestock (Indicator)   | 0.47           | 0.69        | -0.47              | 0.80                  |
| Livestock Heads             | 1.88           | 3.71        | -0.46              | 0.85                  |
| Normalized Asset Index      | 0.02           | 0.41        | -0.43              | .                     |
| N                           | 5,566          | 8,975       |                    |                       |

*Notes:* This table presents summary statistics for refugee and host households in the RIMA sample. Columns (1) and (2) report sample means. Column (3) reports the normalized difference. Column (4) presents reduced-form estimates of the government land allocation (in acres) required for refugee households to achieve parity with host households. Parity allocations are only reported for outcomes with statistically significant coefficients in Table S1. The Coping Strategies Index reports a weighted measure of food security within the past 7 days, with a higher number indicating worse food security outcomes.

Table 3 details several scenarios. Column (1) reports the additional land required to bring every refugee household below a 0.25-acre allocation to the stated policy floor. This ranges from 1.24 to 34.02 percent of the land that is or could be converted to agriculture. These estimates are based on satellite image-based classifications of vegetated land within refugee settlements (19). Columns (2) and (3) calculate the amount of additional land allocation required to achieve parity with hosts,

based upon the estimates in the previous table and the settlement *mean* allocation. At the maximum parity allocation (1.84 acres, for food security), four settlements —Adjumani, Kiryandongo, Kyaka II and Rwamwanja —would require more land than physically available within their respective boundaries. In 2020, these settlements housed 44.8 percent of the total refugee population. This implies that for just under half of refugees, parity through allocation alone is unattainable regardless of political feasibility. The remaining five settlements could in principle accommodate parity through allocations for all outcomes.

However, the feasibility of this rests on particularly strong assumptions: stagnant settlement populations, constant agricultural returns to additional land and uniform land quality. Each of these is unlikely to hold. Settlement populations continue to rise (12), our marginal effects are estimated at the average allocation and likely diminish, and remaining land is likely to be disproportionately marginal. Refugee households interviewed for this study frequently reported their dissatisfaction with the quality of the land that they farm. Because the allocation system relies on agreements where hosts set aside land for refugees (17), the provisioned parcels may often be of limited agricultural quality, and expansion of this land is likely to capture areas of increasingly marginal quality. For these reasons, the feasibility implied by Table 3 should be read as a best case; the true requirements are likely larger and the achievable gains smaller.



Table 3: Net Land Conversion Required for Parity (% of Settlement Area)

| Settlement  | (1)<br>0.25 Acre<br>Baseline | (2)<br>Minimum<br>Expansion | (3)<br>Maximum<br>Expansion |
|-------------|------------------------------|-----------------------------|-----------------------------|
| Adjumani    | 34.02                        | 215.94                      | 607.91                      |
| Bidibidi    | 1.24                         | 9.68                        | 26.27                       |
| Imvepi      | 4.64                         | 20.00                       | 74.36                       |
| Kiryandongo | 19.82                        | 71.99                       | 282.74                      |
| Kyaka II    | 18.49                        | 0.00                        | 332.92                      |
| Palabek     | 2.26                         | 15.04                       | 39.87                       |
| Palorinya   | 6.26                         | 28.14                       | 69.74                       |
| Rhino       | 1.98                         | 9.29                        | 27.49                       |
| Rwamwanja   | 8.40                         | 50.53                       | 234.44                      |

*Notes:* This table reports the net additional land required for refugee households to achieve parity with host households, expressed as a percentage of the available land within each settlement. Available arable land is computed from the 2020 ESA WorldCover (v100, 10m resolution) (19) as the total settlement area net of existing cropland, built-up, water and wetland cover. The first column calculates the total land conversion required to bring all households with sub-0.25 acre allocations up to the 0.25-acre policy goal, assuming no redistribution. Columns 2 and 3 report, respectively, the minimum and maximum estimated land conversion required to raise the settlement mean allocation to a parity target (from Table 2). Refugee populations are UNHCR settlement-level figures as of 31 December 2020.

The analysis that we present here demonstrates that refugees who receive land allocations from the Ugandan government upon arrival fare better than those who do not, and that they are able to leverage this generosity into greater market participation. There appears to be a limit to this, however, as even over a number of years, they do not have significantly higher assets than refugees who do not receive such allocations. Therefore, the policy as it stands supports greater food security, but not necessarily a path out of poverty. In interviews, the refugees themselves suggest alternatives that may be possible to achieve with alternatives to land allocation, such as small loans: many interviewees reported their desire to pivot towards non-agricultural livelihoods, describing the need for small loans in order to “Start a shop instead”, or “Return to schooling” (17).

We note an important limitation to this analysis. We can only make comparisons between refugees in the same setting in Uganda. We are not able to make comparisons between refugees in Uganda and those who are living in situations where there is no such policy. This is also a strength in that we know that we are making comparisons that allow us to uncover an effect that is purged of bias that might occur by comparing refugees in very different situations.

Our findings demonstrate that the policy has important food security effects on refugee households. However, the policy may not be a panacea, since it is limited in its ability to elevate refugees out of poverty. To bring them up to the level of hosts, who are also quite poor, would require substantially more land than is available. Further, given their varied backgrounds, many refugees may not be well-equipped to produce significant agricultural production.

## **5 Materials and methods**

### **5.1 Data Description**

The data source used in this analysis is the Resilience Index Measurement and Analysis (RIMA) survey for Uganda. RIMA is a generalized tool applied by the FAO to estimate household food security. The instruments for each wave can be found online. In Uganda, the sample includes both refugees and hosts. This is an unbalanced panel of data with waves in 2017/18/19, 2019, 2020 and

2021 across 12 settlements, and is designed to be representative of refugee-hosting areas. The first wave is staggered over three years, so the time step between the first and second wave can vary from less than one year to up to 2 years. Subsequent waves are annual, although some households may leave the sample, and the survey adds new households in every wave. We present summary tables describing the average characteristics of households in the analysis sample by the number of waves that they are in (Table S4) and the difference between households that are present in the survey for multiple rounds and those that leave (Table S5). Refugees who stay in the sample for multiple waves tend to have most crops and have higher income. However, there are no large differences in land allocation across these groups, or between stayers and leavers. Those who are in 3 or 4 waves of the survey tend to have higher cash aid (Table S4).

Our analysis is supported by qualitative evidence from interviews conducted across four settlements in Uganda in September 2025. These interviews consist of recorded conversations with four settlement commandants (responsible for settlement management), four appointed refugee community leaders, and 25 refugee household heads. The primary focus of these discussions was to understand and anecdotally corroborate the decision-making processes regarding household production, food security outcomes, local market characteristics, and settlement-level policy (17). The instrument for these interviews is included in the SI section D.

## 5.2 Estimation strategy

We are interested in estimating the effect of agricultural land use rights  $L$  on refugee outcomes  $Y$  for household  $i$  in settlement  $s$  during survey wave  $t$ :

$$Y_{ist} = \alpha + \delta L_{is0} + \mathbf{X}_{ist}\beta t + \phi_{st} + \varepsilon_{ist} \quad (1)$$

$Y$  includes household level measurements of assets, food security, and agricultural production outcomes.  $L_{is0}$  is either the declared size of government allocated parcels or an indicator for having a parcel over 0.25 acres, established in the wave 1 survey. Control variables  $X_{ist}$  at the household

level include household size, sex of respondent, maximum level of schooling of the household, if the household has experienced flooding, drought, crop pests, livestock pests, or high input prices in the past year, as well as indicators for the number of surveys in which the household appears within the panel. Further, we include the number of months since arrival as declared in the baseline survey. This matrix also includes settlement-level measures of maximum temperatures as well as total precipitation over the previous six months. Settlement by year ( $\phi_{st}$ ) fixed effects are included as well. Households which are missing observations for particular variables (such as household size) have the value of that variable replaced with the mean value in their respective settlement, we include an additional indicator in regression equal to 1 if any variable has been replaced with a mean value. We cluster the error term  $\varepsilon$  at the household level.

This specification allows us to use all available household observations, including those that do not appear in all four survey waves. However, because we have the baseline land allocation only for the first year of the survey, we cannot use household fixed effects and must instead exploit variation within settlements in parcel size allocation. In order to circumvent the potential endogeneity and measurement issues of the initial government land allocation discussed in the main text, we use an instrumental variables approach based on a “Leave-One-Out” strategy at the arrival cohort-settlement level. This follows from previous Hausman and Bartik style instruments used to measure market-power (20) and supply-shocks (21). We instrument the individual’s land allocation using the average land area allocated to *other* refugees arriving in the same year (cohort) and settlement as the household, excluding the household itself. Specifically, we estimate a first stage equation that regresses the household’s allocated land on the instrument and controls:

$$L_{is0} = \alpha_0 + \alpha_1 \bar{L}_{s,-i} + \mathbf{X}_{ist} \Gamma + \delta_{st} + \mu_{ist} \quad (2)$$

$L_{is0}$  is the household’s initial government-allocated land size, which is time-invariant for the household. This is instrumented by the average land area allocated to refugees arriving in the same year (cohort) and settlement as the household, excluding the household’s own land allocation  $\bar{L}_{s,-i}$ . Control variables, in the vector  $\mathbf{X}_{ist}$  and fixed effects at the settlement year level ( $\delta_{st}$ ) are the

same as in the second stage. We estimate the system using 2SLS and cluster standard errors at the household level.

The identification strategy relies on the assumption that, conditional on controls, cohort-level land allocations influence household outcomes only through the household's own individual land allocation. We include settlement, year and settlement-year fixed effects, which absorb both invariant and time-varying settlement shocks and fluctuations in aggregate production, average land allocations and local food prices. There is potential concern regarding within-cohort spillovers, such as food sharing from higher allocation households to households perceived as food insecure within the same cohort. Such spillovers would bias the estimated effect of land allocation toward zero. However, we observe the opposite pattern in our main results, suggesting that any such sharing has limited influence on average food security at the settlement level.

Further, one of the elements of the coping strategies index constructed from our data is “borrow food, or rely on help from a friend or neighbor”. The elements of the CSI ask households to report with what frequency they have engaged in a particular strategy during the past 7 days. Borrowing food is the rarest event in our data, and is uncorrelated with our instrument. Specifically, the mean value of the borrowing variable is 0.85, relative to the other available coping strategies, which range from 2.41 to 3.65 days. Further, the raw correlation between our instrument and the borrowing variable is 0.04. We formally test this by estimating the impact of the cohort land allocation on the household's propensity to borrow food over a week and find no significant effect (See SI section A).

Table S6 shows the regression of land allocation on the instrument. The first column contains no covariates, the second household level covariates, and the third the full set of covariates and fixed effects that are present in the second stage regression. A greater cohort-average land allocation strongly predicts household land allocations, and the F-statistic of the instrument is around 38 even with the full set of controls.

We implement a number of robustness checks. These include decomposing the effect of land allocation on individual components of the Coping Strategies Index, estimating the impact of hav-

ing more than 0.25 acres of land (an indicator rather than a continuous treatment), and examining various subsamples of the survey (the full panel, those present in 2 waves, etc). These results are discussed in more detail in the SI section A.

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## **Supplementary Information**

### ***Cultivating self-reliance? An analysis of Uganda's refugee land allocation policy***

#### **Table of Contents**

|          |                                                   |           |
|----------|---------------------------------------------------|-----------|
| <b>A</b> | <b>Discussion of additional robustness checks</b> | <b>2</b>  |
| <b>B</b> | <b>Supplemental tables</b>                        | <b>3</b>  |
| <b>C</b> | <b>Supplemental figures</b>                       | <b>16</b> |
| <b>D</b> | <b>Structured interview instrument</b>            | <b>19</b> |

## A Discussion of additional robustness checks

Tables S7 through S13 present a series of robustness checks. We first decompose the effect of land allocation on the individual components of the Coping Strategies Index (Table S7). Improved food security from land allocation is driven primarily by reductions in dietary adjustments, specifically the consumption of less preferred foods and the frequency of skipping meals. We detect no effect of land allocation on distress sales, consumption of seed stock, financial burdening or borrowing from peers. The null result on borrowing is particularly notable, as it suggests that land-rich households are not acting as lenders or food sources for neighbors, supporting the validity of our exclusion restriction regarding spillovers. This is further supported by a more restrictive test in Table S12. We estimate the impact of allocation on individual coping strategies for households with land allocations below their cohort average. If our concerns of exclusion restriction violation were actualized, this group would be the most likely to rely on their cohort peers for ways to maintain food security/earnings. Notably, we find that our individual coping strategies again are primarily driven by reductions in dietary adjustments and the number of skipped meals.

Next, we test the effect of the specific policy threshold: having a land allocation of at least 0.25 acres (Table S8). The direction and magnitude of estimates remain similar to our main specifications, though we lose statistical significance for the extensive margin of earnings as well as the intensive margins of earnings and livestock count for households who report positive amounts of either measure. Notably, we see a significant *reduction* in households who report positive harvests. This may be driven by higher endowments leading to non-agricultural incomes.

To address concerns of potential bias arising from study attrition, we test if land allocation influences the decision to leave the settlement (and therefore the survey). We regress an indicator for attrition on initial land allocation, finding no significant relationship between land size and the decision to leave across our preferred specifications (Table S9). A comparison of observable characteristics between households who stay and do not stay in the survey (Table S5) reveals no significant discrepancies, with the minor exception of cash (voucher) aid received. The lack of discrepancy in settlement attrition is likely driven by the aid policy structure: while refugees possess freedom of movement, leaving the settlement requires relinquishing both the allocated land and access to direct food and cash aid (3).

We also examine impacts for households present in at least two survey waves and for the fully balanced panel (Tables S10 and S11). For households present in at least two waves we find similar results to our main specification, with a positive and marginally significant effect ( $p < 0.1$ ) on asset accumulation. For households present in all waves, we find similar results to our main specification though we do find insignificant results for the intensive margins of livestock, for households who report a positive number of heads. We report the characteristics of households by the number of waves they are present in (Table S4). We find that households present in all waves of the survey are likely to have higher land allocations, more earnings, more acreage of crops planted, lower amounts of food aid (offset by higher amounts of cash aid) and longer tenures within the settlement. Finally, we restrict the sample to households who do not have missing data for any of our controls. We run estimations on a fully non-imputed sample in Table S13. We find similar results to our main estimates, with the notable exception of insignificant coefficients for food security and the intensive margin of earnings. Nonetheless, these coefficients do maintain the same magnitude in these estimates, suggesting that the discrepancy may be due to reduced power from the smaller sample size.

## **B Supplemental tables**

Table S1: Leave-One-Out Results: Food Security and Production

|                                    | (1)<br>Food Security | (2)<br>Any Harvest  | (3)<br>ln(Harvest > 0) | (4)<br>Any Earnings | (5)<br>ln(Earnings > 0) | (6)<br>ln(Land Use) | (7)<br>Any Livestock | (8)<br>ln(Livestock > 0) | (9)<br>Asset Index |
|------------------------------------|----------------------|---------------------|------------------------|---------------------|-------------------------|---------------------|----------------------|--------------------------|--------------------|
| <b>Panel A: IV (Leave-One-Out)</b> |                      |                     |                        |                     |                         |                     |                      |                          |                    |
| Gov't Land                         | 0.270**<br>(0.125)   | -0.062<br>(0.042)   | 0.729***<br>(0.236)    | 0.204***<br>(0.070) | 0.852**<br>(0.378)      | 0.806***<br>(0.232) | 0.282***<br>(0.070)  | 0.332**<br>(0.162)       | 0.207<br>(0.138)   |
| Mean Dep. Var.                     | 0.06                 | 0.86                | 4.47                   | 0.30                | 2.76                    | -1.25               | 0.47                 | 0.99                     | 0.02               |
| Obs.                               | 5560                 | 5566                | 4596                   | 5566                | 1475                    | 4772                | 5566                 | 2590                     | 5566               |
| R-sq                               | 0.076                | 0.149               | 0.422                  | 0.154               | 0.247                   | 0.269               | 0.216                | 0.460                    | 0.175              |
| WeakID (Wald F)                    | 37.79                | 37.76               | 38.03                  | 37.76               | 20.51                   | 38.86               | 37.76                | 26.00                    | 37.76              |
| UnderID P-Val                      | 0.000                | 0.000               | 0.000                  | 0.000               | 0.002                   | 0.000               | 0.000                | 0.000                    | 0.000              |
| <b>Panel B: Naive OLS</b>          |                      |                     |                        |                     |                         |                     |                      |                          |                    |
| Gov't Land                         | 0.070**<br>(0.029)   | 0.040***<br>(0.010) | 0.195***<br>(0.049)    | 0.075***<br>(0.015) | 0.076<br>(0.085)        | 0.449***<br>(0.058) | 0.053***<br>(0.015)  | -0.044<br>(0.030)        | 0.029<br>(0.033)   |
| Mean Dep. Var.                     | 0.06                 | 0.86                | 4.47                   | 0.30                | 2.76                    | -1.25               | 0.47                 | 0.99                     | 0.02               |
| Obs.                               | 5560                 | 5566                | 4596                   | 5566                | 1475                    | 4772                | 5566                 | 2590                     | 5566               |
| R-sq                               | 0.084                | 0.165               | 0.442                  | 0.169               | 0.288                   | 0.277               | 0.254                | 0.492                    | 0.183              |
| Covariates                         | Yes                  | Yes                 | Yes                    | Yes                 | Yes                     | Yes                 | Yes                  | Yes                      | Yes                |
| Settlement/Year FE                 | Yes                  | Yes                 | Yes                    | Yes                 | Yes                     | Yes                 | Yes                  | Yes                      | Yes                |

*Notes:* Panel A presents coefficients from an instrumental variables regression using the instrument described in the Materials and Methods section, while panel B presents the same estimates using OLS. Each column represents a separate regression with dependent variables as listed in the column headers. The estimators include fixed effects at the settlement level, year effects, and settlement by year effects. Household controls include household size, sex of respondent, maximum level of schooling of the household, if the household has experienced flooding, drought, crop pests, livestock pests, or high input prices in the past year, number of months since arrival as declared in the baseline survey, as well as indicators for the number of surveys in which the household appears within the panel. Settlement level measures of maximum temperatures as well as total precipitation over the previous six months are also included. Households which are missing observations for particular variables (such as household size) have the value of that variable replaced with the mean value in their respective settlement, we include an additional indicator in regression equal to 1 if such a substitution has been made. Standard errors are clustered at the household level in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Table S2: Leave-One-Out Results: Livestock Outcomes

|                                    | (1)<br>Heads of Livestock | (2)<br>IHS(Sales)   | (3)<br>IHS(Income) |
|------------------------------------|---------------------------|---------------------|--------------------|
| <b>Panel A: IV (Leave-One-Out)</b> |                           |                     |                    |
| Gov't Land                         | 1.209<br>(0.744)          | 0.382***<br>(0.140) | 1.293**<br>(0.592) |
| Mean Dep. Var.                     | 4.04                      | 0.25                | 1.29               |
| Obs.                               | 2590                      | 5566                | 5566               |
| R-sq                               | 0.263                     | 0.038               | 0.077              |
| WeakID (Wald F)                    | 26.00                     | 37.76               | 37.76              |
| UnderID P-Val                      | 0.000                     | 0.000               | 0.000              |
| <b>Panel B: Naive OLS</b>          |                           |                     |                    |
| Gov't Land                         | -0.123<br>(0.150)         | 0.026<br>(0.026)    | 0.099<br>(0.112)   |
| Mean Dep. Var.                     | 4.04                      | 0.25                | 1.29               |
| Obs.                               | 2590                      | 5566                | 5566               |
| R-sq                               | 0.284                     | 0.069               | 0.095              |
| Covariates                         | Yes                       | Yes                 | Yes                |
| Settlement/Year FE                 | Yes                       | Yes                 | Yes                |

*Notes:* Panel A presents coefficients from an instrumental variables regression using the instrument described in the Materials and Methods section, while panel B presents the same estimates using OLS. Each column represents a separate regression with dependent variables as listed in the column headers. Sales and income are transformed using the inverse hyperbolic sine (IHS) to retain zero values and preserve statistical power. Estimates include the full set of household controls, weather controls, and fixed effects described in Table S1. Standard errors are clustered at the household level.

Table S3: Impact Heterogeneity: Early vs Late (Pooled Interaction)

|                                | (1)               | (2)               | (3)              | (4)                | (5)               | (6)              | (7)                | (8)                | (9)               |
|--------------------------------|-------------------|-------------------|------------------|--------------------|-------------------|------------------|--------------------|--------------------|-------------------|
|                                | Food Security     | Any Harvest       | ln(Harvest > 0)  | Any Earnings       | ln(Earnings > 0)  | ln(Land Use)     | Any Livestock      | ln(Livestock > 0)  | Asset Index       |
| Gov't Land (Early Effect)      | 0.285<br>(0.221)  | -0.139<br>(0.098) | 0.334<br>(0.487) | -0.026<br>(0.124)  | 1.121<br>(0.876)  | 0.474<br>(0.564) | 0.300**<br>(0.145) | 0.561**<br>(0.284) | -0.130<br>(0.247) |
| Gov't Land × Late (Difference) | -0.016<br>(0.175) | 0.076<br>(0.086)  | 0.382<br>(0.363) | 0.225**<br>(0.094) | -0.276<br>(0.653) | 0.321<br>(0.432) | -0.018<br>(0.114)  | -0.224<br>(0.189)  | 0.331*<br>(0.187) |
| Total Late Effect              | 0.270             | -0.063            | 0.716            | 0.200              | 0.844             | 0.795            | 0.282              | 0.336              | 0.201             |
| Late P-Val                     | 0.032             | 0.131             | 0.003            | 0.004              | 0.029             | 0.001            | 0.000              | 0.040              | 0.149             |
| Mean Dep. Var.                 | 0.06              | 0.86              | 4.47             | 0.30               | 2.76              | -1.25            | 0.47               | 0.99               | 0.02              |
| Obs.                           | 5560              | 5566              | 4596             | 5566               | 1475              | 4772             | 5566               | 2590               | 5566              |
| WeakID (Wald F)                | 18.47             | 18.46             | 16.85            | 18.46              | 9.02              | 17.34            | 18.46              | 12.20              | 18.46             |
| UnderID P-Val                  | 0.000             | 0.000             | 0.000            | 0.000              | 0.002             | 0.000            | 0.000              | 0.000              | 0.000             |

Notes: The table presents coefficients from an instrumental variables regression using the instrument described in the Materials and Methods section. Each column represents a separate regression with dependent variables as listed in the column headers. Row 1 shows the effect for years  $\leq 2019$ . Row 2 shows the differential for years  $> 2019$ . Households with at least two observations are included in the sample. Total Late Effect is the sum of the two coefficients. Estimates include the full set of household controls, weather controls, and fixed effects described in Table S1. Standard errors are clustered at the household level.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$



Table S4: Refugee Household Characteristics by Number of Surveys in Panel

| Outcome Variable                  | Number of Surveys in Panel |         |         |         |
|-----------------------------------|----------------------------|---------|---------|---------|
|                                   | 1                          | 2       | 3       | 4       |
| <i>Land allocation</i>            |                            |         |         |         |
| Allocated $\geq 0.25$ acres       | 0.38                       | 0.35    | 0.36    | 0.44    |
| Gov't Land Allocation (Acres)     | 0.39                       | 0.34    | 0.35    | 0.40    |
| <i>Household characteristics</i>  |                            |         |         |         |
| Arrival Year                      | 2016.77                    | 2015.17 | 2016.93 | 2013.89 |
| Household Size                    | 5.34                       | 5.89    | 6.20    | 6.68    |
| Male HH Head                      | 0.60                       | 0.57    | 0.60    | 0.59    |
| Years of School                   | 8.92                       | 7.90    | 8.28    | 8.24    |
| Farming HH                        | 0.47                       | 0.45    | 0.52    | 0.56    |
| Pastoralist HH                    | 0.11                       | 0.11    | 0.10    | 0.13    |
| Cash Aid (USD), 12 mos            | 71.38                      | 91.43   | 101.18  | 156.12  |
| In-Kind Aid (USD), 12 mos         | 159.28                     | 155.83  | 178.19  | 145.46  |
| <i>Outcomes</i>                   |                            |         |         |         |
| Coping Strategies Index           | 14.97                      | 15.31   | 14.75   | 14.60   |
| Total Crop Harvest (KG), 12 mos   | 107.71                     | 125.16  | 165.33  | 231.16  |
| Total Crop Sales (KG), 12 mos     | 25.03                      | 26.39   | 35.07   | 56.03   |
| Total Crop Earnings (USD), 12 mos | 6.89                       | 7.48    | 9.14    | 13.76   |
| Crop Earnings Ind.                | 0.43                       | 0.34    | 0.32    | 0.34    |
| Total Land Use (Acres)            | 0.58                       | 0.54    | 0.59    | 0.71    |
| Normalized Asset Index            | -0.32                      | -0.32   | -0.15   | -0.17   |
| Heads of Livestock Owned          | 1.93                       | 1.51    | 1.95    | 2.15    |
| Livestock Earnings (USD), 12 mos  | 3.16                       | 3.43    | 4.20    | 5.80    |
| Observations                      | 1,353                      | 1,325   | 2,604   | 3,248   |

*Notes:* The table compares the mean household characteristics by the number of RIMA survey rounds (2017-2021) in which the household is present. Columns (1)-(4) indicate the number of rounds the household is observed.

Table S5: Household Characteristics by Household Presence

| Outcome Variable                  | Stayers | Left Survey | Norm. Diff. |
|-----------------------------------|---------|-------------|-------------|
| <i>Land allocation</i>            |         |             |             |
| Allocated $\geq 0.25$ acres       | 0.41    | 0.37        | -0.08       |
| Gov't Land Allocation (Acres)     | 0.38    | 0.36        | -0.06       |
| <i>Household characteristics</i>  |         |             |             |
| Arrival Year                      | 2015.78 | 2016.77     | 0.29        |
| Household Size                    | 6.38    | 5.87        | -0.16       |
| Male HH Head                      | 0.57    | 0.63        | 0.11        |
| Years of School                   | 8.54    | 7.87        | -0.16       |
| Farming HH                        | 0.47    | 0.61        | 0.30        |
| Pastoralist HH                    | 0.13    | 0.09        | -0.14       |
| Cash Aid (USD), 12 mos            | 122.68  | 114.76      | -0.04       |
| In-Kind Aid (USD), 12 mos         | 146.83  | 177.62      | 0.16        |
| <i>Outcomes</i>                   |         |             |             |
| Coping Strategies Index           | 15.18   | 14.13       | -0.09       |
| Total Crop Harvest (KG), 12 mos   | 195.70  | 145.43      | -0.15       |
| Total Crop Sales (KG), 12 mos     | 46.30   | 30.82       | -0.12       |
| Total Crop Earnings (USD), 12 mos | 11.32   | 8.83        | -0.07       |
| Crop Earnings Ind.                | 0.36    | 0.33        | -0.06       |
| Total Land Use (Acres)            | 0.64    | 0.60        | -0.04       |
| Normalized Asset Index            | -0.18   | -0.28       | -0.11       |
| Heads of Livestock Owned          | 2.12    | 1.65        | -0.14       |
| Livestock Earnings (USD), 12 mos  | 4.81    | 4.00        | -0.05       |
| Observations                      | 5,544   | 2,986       |             |

*Notes:* The table compares characteristics of refugee households that remained in the panel (Stayers) versus those that were not present for all four waves (Left Survey). column (3) reports the normalized difference in means.

Table S6: First Stage: Effect of Instrument on Household Land Allocation

|                              | (1)<br>No Controls    | (2)<br>Main Controls  | (3)<br>Controls + FE  |
|------------------------------|-----------------------|-----------------------|-----------------------|
| Leave-One-Out Avg. Land (IV) | 0.9350***<br>(0.0589) | 0.8425***<br>(0.0855) | 0.7213***<br>(0.1174) |
| Observations                 | 5566                  | 5566                  | 5566                  |
| R-Squared                    | 0.26                  | 0.27                  | 0.28                  |
| First Stage F-Stat           | 251.64                | 97.15                 | 37.76                 |
| Covariates                   | No                    | Yes                   | Yes                   |
| Settlement/Year FE           | No                    | No                    | Yes                   |

*Notes:* This table reports first-stage estimates of the effect of the leave-one-out average government land allocation instrument—described in the Materials and Methods section—on household-level allocated land area (in acres). Column (1) presents the unconditional bivariate regression. Column (2) adds household and weather controls (household size, gender of head, years of schooling, indicator for no schooling, number of survey waves, log months in settlement, relief aid received, imputation indicator, six-month total precipitation, six-month maximum temperature, and weather interactions). Column (3) further includes settlement, year and settlement-by-year fixed effects. Standard errors clustered at the household level are reported in parentheses. The first-stage F-statistic tests the joint significance of the excluded instrument.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Table S7: Leave-One-Out Results: Individual Coping Strategies

|                                         | (1)<br>Less Pref.   | (2)<br>Limit Size | (3)<br>Red. Meals  | (4)<br>Restr. Adult | (5)<br>Red. Other | (6)<br>Skip Day     |
|-----------------------------------------|---------------------|-------------------|--------------------|---------------------|-------------------|---------------------|
| <b>Panel A: Dietary Adjustments</b>     |                     |                   |                    |                     |                   |                     |
| Gov't Land                              | -0.959**<br>(0.386) | -0.439<br>(0.332) | -0.566*<br>(0.296) | -0.228<br>(0.204)   | -0.097<br>(0.129) | -0.282**<br>(0.128) |
| Mean Dep. Var.                          | 3.52                | 2.60              | 2.26               | 0.99                | 0.27              | 0.38                |
| Obs.                                    | 5565                | 5566              | 5564               | 5565                | 5565              | 5564                |
| R-sq                                    | 0.079               | 0.076             | 0.080              | 0.057               | 0.079             | 0.049               |
| WeakID (Wald F)                         | 37.74               | 37.74             | 37.74              | 37.69               | 36.56             | 37.77               |
| UnderID P-Val                           | 0.000               | 0.000             | 0.000              | 0.000               | 0.000             | 0.000               |
| <b>Panel B: Livelihood &amp; Assets</b> |                     |                   |                    |                     |                   |                     |
|                                         | Immature            | Seed Stock        | Nat. Res.          | Sell Assets         | Barter            | Work 4 Food         |
| Gov't Land                              | 0.064<br>(0.157)    | -0.013<br>(0.196) | -0.097<br>(0.090)  | -0.035<br>(0.075)   | 0.091<br>(0.165)  | 0.080<br>(0.194)    |
| Mean Dep. Var.                          | 0.50                | 0.53              | 0.27               | 0.13                | 0.60              | 0.54                |
| Obs.                                    | 5565                | 5565              | 5566               | 5566                | 5566              | 5564                |
| R-sq                                    | 0.081               | 0.074             | 0.104              | 0.046               | 0.077             | 0.078               |
| WeakID (Wald F)                         | 37.72               | 37.74             | 37.74              | 37.74               | 37.74             | 37.76               |
| UnderID P-Val                           | 0.000               | 0.000             | 0.000              | 0.000               | 0.000             | 0.000               |
| <b>Panel C: Social &amp; Extreme</b>    |                     |                   |                    |                     |                   |                     |
|                                         | Credit              | Borrow            | Wild Food          | Send Away           | Beg               | School Food         |
| Gov't Land                              | -0.071<br>(0.175)   | -0.042<br>(0.165) | 0.016<br>(0.141)   | -0.137*<br>(0.082)  | -0.082<br>(0.111) | -0.018<br>(0.061)   |
| Mean Dep. Var.                          | 0.79                | 0.81              | 0.44               | 0.17                | 0.36              | 0.08                |
| Obs.                                    | 5562                | 5564              | 5565               | 5564                | 5564              | 5566                |
| R-sq                                    | 0.093               | 0.046             | 0.095              | 0.028               | 0.049             | 0.044               |
| WeakID (Wald F)                         | 37.83               | 37.82             | 37.74              | 37.72               | 37.73             | 37.74               |
| UnderID P-Val                           | 0.000               | 0.000             | 0.000              | 0.000               | 0.000             | 0.000               |
| Covariates                              | Yes                 | Yes               | Yes                | Yes                 | Yes               | Yes                 |
| Settlement/Year FE                      | Yes                 | Yes               | Yes                | Yes                 | Yes               | Yes                 |

*Notes:* The table presents coefficients from an instrumental variables regression using the instrument described in the Materials and Methods section. All outcomes represent the number of days in the past 7 days the household employed the coping strategy. Estimates include the full set of household controls, weather controls, and fixed effects described in Table S1. Standard errors are clustered at the household level in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Panel A (Dietary):** *Less Pref.* = Rely on less preferred/expensive foods; *Limit Size* = Limit portion size at mealtimes; *Red. Meals* = Reduce number of meals eaten in a day; *Restr. Adult* = Restrict adult consumption for children; *Red. Other* = Reduce consumption by others so workers can eat; *Skip Day* = Go an entire day without eating.

**Panel B (Livelihood):** *Immature* = Harvest immature crops; *Seed Stock* = Consume seed stock intended for next season; *Nat. Res.* = Sell natural resources (charcoal, firewood, etc.); *Sell Assets* = Sell household assets; *Barter* = Exchange goods for food; *Work 4 Food* = Exchange labor for food.

**Panel C (Social):** *Credit* = Purchase food on credit; *Borrow* = Borrow food from friends/relatives; *Wild Food* = Gather wild food or hunt; *Send Away* = Send household member to eat elsewhere; *Beg* = Beg for food; *School Food* = Send children to school specifically for meals.

Table S8: Leave-One-Out and Naive Results: Household's w/  $\geq 0.25$  Acre Allocation

|                                    | (1)                | (2)                 | (3)                | (4)                | (5)               | (6)                 | (7)                 | (8)               | (9)              |
|------------------------------------|--------------------|---------------------|--------------------|--------------------|-------------------|---------------------|---------------------|-------------------|------------------|
|                                    | Food Security      | Any Harvest         | ln(Harvest > 0)    | Any Earnings       | ln(Earnings > 0)  | ln(Land Use)        | Any Livestock       | ln(Livestock > 0) | Asset Index      |
| <b>Panel A: IV (Leave-One-Out)</b> |                    |                     |                    |                    |                   |                     |                     |                   |                  |
| Gov't Land                         | 0.304**<br>(0.141) | -0.101**<br>(0.048) | 0.517**<br>(0.238) | 0.128<br>(0.086)   | 0.474<br>(0.333)  | 0.417*<br>(0.235)   | 0.299***<br>(0.075) | 0.251<br>(0.160)  | 0.200<br>(0.164) |
| Mean Dep. Var.                     | 0.08               | 0.89                | 4.99               | 0.41               | 3.07              | -0.76               | 0.48                | 0.93              | 0.12             |
| Obs.                               | 2200               | 2201                | 1893               | 2201               | 838               | 1934                | 2201                | 1065              | 2201             |
| R-sq                               | 0.059              | 0.208               | 0.456              | 0.199              | 0.251             | 0.217               | 0.192               | 0.507             | 0.161            |
| WeakID (Wald F)                    | 35.25              | 35.25               | 34.37              | 35.25              | 28.63             | 34.21               | 35.25               | 27.33             | 35.25            |
| UnderID P-Val                      | 0.000              | 0.000               | 0.000              | 0.000              | 0.002             | 0.000               | 0.000               | 0.000             | 0.000            |
| <b>Panel B: Naive OLS</b>          |                    |                     |                    |                    |                   |                     |                     |                   |                  |
| Gov't Land                         | 0.082**<br>(0.036) | 0.001<br>(0.012)    | 0.124**<br>(0.059) | 0.046**<br>(0.020) | 0.184*<br>(0.107) | 0.334***<br>(0.071) | 0.053***<br>(0.019) | -0.025<br>(0.037) | 0.056<br>(0.044) |
| Mean Dep. Var.                     | 0.08               | 0.89                | 4.99               | 0.41               | 3.07              | -0.76               | 0.48                | 0.93              | 0.12             |
| Obs.                               | 2200               | 2201                | 1893               | 2201               | 838               | 1934                | 2201                | 1065              | 2201             |
| R-sq                               | 0.075              | 0.237               | 0.473              | 0.207              | 0.258             | 0.218               | 0.259               | 0.532             | 0.168            |
| Covariates                         | Yes                | Yes                 | Yes                | Yes                | Yes               | Yes                 | Yes                 | Yes               | Yes              |
| Settlement/Year FE                 | Yes                | Yes                 | Yes                | Yes                | Yes               | Yes                 | Yes                 | Yes               | Yes              |

Notes: Panel A presents coefficients from an instrumental variables regression using the instrument described in the Materials and Methods section, while panel B presents the same estimates using OLS. These estimates are restricted to the sub-sample of households with an initial government land allocation greater than or equal to 0.25 acres. Each column represents a separate regression with dependent variables as listed in the column headers. Both panels include the full set of household controls, weather controls, and fixed effects described in Table S1. Standard errors are clustered at the household level.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Table S9: Effect of Government Land Allocation on Refugee Household Panel Attrition

|                    | (1)<br>No Controls       | (2)<br>Controls       | (3)<br>TWFE          | (4)<br>TWFE-IV        |
|--------------------|--------------------------|-----------------------|----------------------|-----------------------|
| Gov't Land (Acres) | -0.02448***<br>(0.00830) | -0.00274<br>(0.01079) | 0.00630<br>(0.00747) | -0.02430<br>(0.03216) |
| Mean Dep. Var.     | 0.280                    | 0.280                 | 0.280                | 0.280                 |
| Observations       | 5566                     | 5566                  | 5566                 | 5566                  |
| R-squared          | 0.001                    | 0.356                 | 0.827                | 0.827                 |
| WeakID (Wald F)    |                          |                       |                      | 37.76                 |
| UnderID P-Val      |                          |                       |                      | 0.000                 |
| Covariates         | No                       | Yes                   | Yes                  | Yes                   |
| Settlement/Year FE | No                       | No                    | Yes                  | Yes                   |
| Instrument         | No                       | No                    | No                   | Yes                   |

*Notes:* This table reports estimates of the effect of government land allocation on the probability that a refugee household was not present for all waves of the RIMA survey. Column (1) reports the unconditional bivariate OLS regression. Column (2) adds household and weather controls. Column (3) further includes settlement, year and settlement-by-year fixed effects. Column (4) instruments household land allocation using the leave-one-out cohort-average allocation. Standard errors are clustered at the household level.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Table S10: Leave-One-Out and Naive Results: Household's Present in at Least 2 Waves

|                                    | (1)<br>Food Security | (2)<br>Any Harvest  | (3)<br>ln(Harvest > 0) | (4)<br>Any Earnings | (5)<br>ln(Earnings > 0) | (6)<br>ln(Land Use) | (7)<br>Any Livestock | (8)<br>ln(Livestock > 0) | (9)<br>Asset Index |
|------------------------------------|----------------------|---------------------|------------------------|---------------------|-------------------------|---------------------|----------------------|--------------------------|--------------------|
| <b>Panel A: IV (Leave-One-Out)</b> |                      |                     |                        |                     |                         |                     |                      |                          |                    |
| Gov't Land                         | 0.262**<br>(0.123)   | -0.037<br>(0.038)   | 0.648***<br>(0.226)    | 0.197***<br>(0.068) | 0.782**<br>(0.359)      | 0.789***<br>(0.227) | 0.277***<br>(0.068)  | 0.323**<br>(0.156)       | 0.238*<br>(0.134)  |
| Mean Dep. Var.                     | 0.07                 | 0.86                | 4.53                   | 0.30                | 2.78                    | -1.20               | 0.48                 | 1.03                     | 0.03               |
| Obs.                               | 5195                 | 5201                | 4304                   | 5201                | 1391                    | 4462                | 5201                 | 2495                     | 5201               |
| R-sq                               | 0.074                | 0.159               | 0.406                  | 0.149               | 0.251                   | 0.245               | 0.218                | 0.437                    | 0.173              |
| WeakID (Wald F)                    | 39.37                | 39.33               | 39.65                  | 39.33               | 22.96                   | 40.55               | 39.33                | 28.66                    | 39.33              |
| UnderID P-Val                      | 0.000                | 0.000               | 0.000                  | 0.000               | 0.002                   | 0.000               | 0.000                | 0.000                    | 0.000              |
| <b>Panel B: Naive OLS</b>          |                      |                     |                        |                     |                         |                     |                      |                          |                    |
| Gov't Land                         | 0.054*<br>(0.031)    | 0.036***<br>(0.010) | 0.188***<br>(0.051)    | 0.069***<br>(0.016) | 0.061<br>(0.091)        | 0.402***<br>(0.060) | 0.044***<br>(0.015)  | -0.048<br>(0.031)        | 0.026<br>(0.034)   |
| Mean Dep. Var.                     | 0.07                 | 0.86                | 4.53                   | 0.30                | 2.78                    | -1.20               | 0.48                 | 1.03                     | 0.03               |
| Obs.                               | 5195                 | 5201                | 4304                   | 5201                | 1391                    | 4462                | 5201                 | 2495                     | 5201               |
| R-sq                               | 0.082                | 0.167               | 0.422                  | 0.163               | 0.284                   | 0.254               | 0.257                | 0.468                    | 0.184              |
| Covariates                         | Yes                  | Yes                 | Yes                    | Yes                 | Yes                     | Yes                 | Yes                  | Yes                      | Yes                |
| Settlement/Year FE                 | Yes                  | Yes                 | Yes                    | Yes                 | Yes                     | Yes                 | Yes                  | Yes                      | Yes                |

*Notes:* Panel A presents coefficients from an instrumental variables regression using the instrument described in the Materials and Methods section, while panel B presents the same estimates using OLS. These estimates are restricted to the sub-sample of households present in at least two survey waves. Each column represents a separate regression with dependent variables as listed in the column headers. Both panels include the full set of household controls, weather controls, and fixed effects described in Table S1. Standard errors are clustered at the household level.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Table S11: Leave-One-Out and Naive Results: Fully Balanced Panel

|                                    | (1)                | (2)                | (3)                | (4)                 | (5)              | (6)                 | (7)                 | (8)               | (9)              |
|------------------------------------|--------------------|--------------------|--------------------|---------------------|------------------|---------------------|---------------------|-------------------|------------------|
|                                    | Food Security      | Any Harvest        | ln(Harvest > 0)    | Any Earnings        | ln(Earnings > 0) | ln(Land Use)        | Any Livestock       | ln(Livestock > 0) | Asset Index      |
| <b>Panel A: IV (Leave-One-Out)</b> |                    |                    |                    |                     |                  |                     |                     |                   |                  |
| Gov't Land                         | 0.279**<br>(0.135) | -0.044<br>(0.044)  | 0.428*<br>(0.241)  | 0.163**<br>(0.080)  | 0.603<br>(0.377) | 0.659***<br>(0.252) | 0.281***<br>(0.076) | 0.149<br>(0.156)  | 0.231<br>(0.165) |
| Mean Dep. Var.                     | 0.08               | 0.88               | 4.75               | 0.34                | 3.00             | -1.08               | 0.52                | 1.10              | 0.03             |
| Obs.                               | 2593               | 2596               | 2184               | 2596                | 763              | 2259                | 2596                | 1352              | 2596             |
| R-sq                               | 0.064              | 0.154              | 0.416              | 0.172               | 0.260            | 0.255               | 0.217               | 0.427             | 0.212            |
| WeakID (Wald F)                    | 27.07              | 27.00              | 33.27              | 27.00               | 22.67            | 34.18               | 27.00               | 26.14             | 27.00            |
| UnderID P-Val                      | 0.000              | 0.000              | 0.000              | 0.000               | 0.008            | 0.000               | 0.000               | 0.000             | 0.000            |
| <b>Panel B: Naive OLS</b>          |                    |                    |                    |                     |                  |                     |                     |                   |                  |
| Gov't Land                         | 0.074*<br>(0.045)  | 0.032**<br>(0.014) | 0.165**<br>(0.069) | 0.085***<br>(0.023) | 0.019<br>(0.100) | 0.274***<br>(0.077) | 0.047**<br>(0.021)  | -0.035<br>(0.043) | 0.043<br>(0.044) |
| Mean Dep. Var.                     | 0.08               | 0.88               | 4.75               | 0.34                | 3.00             | -1.08               | 0.52                | 1.10              | 0.03             |
| Obs.                               | 2593               | 2596               | 2184               | 2596                | 763              | 2259                | 2596                | 1352              | 2596             |
| R-sq                               | 0.073              | 0.165              | 0.421              | 0.177               | 0.288            | 0.265               | 0.258               | 0.435             | 0.221            |
| Covariates                         | Yes                | Yes                | Yes                | Yes                 | Yes              | Yes                 | Yes                 | Yes               | Yes              |
| Settlement/Year FE                 | Yes                | Yes                | Yes                | Yes                 | Yes              | Yes                 | Yes                 | Yes               | Yes              |

Notes: Panel A presents coefficients from an instrumental variables regression using the instrument described in the Materials and Methods section, while panel B presents the same estimates using OLS. These estimates are restricted to the sub-sample of households present in every wave of the survey. Each column represents a separate regression with dependent variables as listed in the column headers. Both panels include the full set of household controls, weather controls, and fixed effects described in Table S1. Standard errors are clustered at the household level.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Table S12: Exclusion Restriction Test: Coping Strategies of Households with Allocations below Cohort Average

|                                         | (1)<br>Less Pref.   | (2)<br>Limit Size | (3)<br>Red. Meals  | (4)<br>Restr. Adult | (5)<br>Red. Other | (6)<br>Skip Day   |
|-----------------------------------------|---------------------|-------------------|--------------------|---------------------|-------------------|-------------------|
| <b>Panel A: Dietary Adjustments</b>     |                     |                   |                    |                     |                   |                   |
| Gov't Land                              | -2.488**<br>(1.084) | -0.391<br>(0.912) | -1.584*<br>(0.934) | 0.186<br>(0.609)    | 0.171<br>(0.321)  | -0.608<br>(0.404) |
| Mean Dep. Var.                          | 3.56                | 2.63              | 2.31               | 1.00                | 0.29              | 0.39              |
| Obs.                                    | 2944                | 2945              | 2945               | 2944                | 2945              | 2945              |
| R-sq                                    | 0.100               | 0.109             | 0.095              | 0.074               | 0.095             | 0.083             |
| WeakID (Wald F)                         | 21.85               | 21.85             | 21.85              | 21.81               | 21.85             | 21.85             |
| UnderID P-Val                           | 0.000               | 0.000             | 0.000              | 0.000               | 0.000             | 0.000             |
| <b>Panel B: Livelihood &amp; Assets</b> |                     |                   |                    |                     |                   |                   |
|                                         | Immature            | Seed Stock        | Nat. Res.          | Sell Assets         | Barter            | Work 4 Food       |
| Gov't Land                              | 0.402<br>(0.443)    | -0.338<br>(0.550) | -0.391<br>(0.286)  | -0.019<br>(0.226)   | -0.132<br>(0.472) | 0.170<br>(0.547)  |
| Mean Dep. Var.                          | 0.52                | 0.56              | 0.30               | 0.14                | 0.59              | 0.55              |
| Obs.                                    | 2944                | 2945              | 2945               | 2945                | 2945              | 2944              |
| R-sq                                    | 0.084               | 0.068             | 0.115              | 0.050               | 0.085             | 0.096             |
| WeakID (Wald F)                         | 21.89               | 21.85             | 21.85              | 21.85               | 21.85             | 21.85             |
| UnderID P-Val                           | 0.000               | 0.000             | 0.000              | 0.000               | 0.000             | 0.000             |
| <b>Panel C: Social &amp; Extreme</b>    |                     |                   |                    |                     |                   |                   |
|                                         | Credit              | Borrow            | Wild Food          | Send Away           | Beg               | School Food       |
| Gov't Land                              | 0.129<br>(0.515)    | 0.675<br>(0.528)  | -0.155<br>(0.408)  | -0.282<br>(0.235)   | -0.111<br>(0.322) | -0.051<br>(0.184) |
| Mean Dep. Var.                          | 0.81                | 0.83              | 0.45               | 0.19                | 0.37              | 0.09              |
| Obs.                                    | 2943                | 2943              | 2944               | 2943                | 2944              | 2945              |
| R-sq                                    | 0.095               | 0.051             | 0.100              | 0.040               | 0.057             | 0.047             |
| WeakID (Wald F)                         | 21.93               | 21.93             | 21.85              | 21.89               | 21.74             | 21.85             |
| UnderID P-Val                           | 0.000               | 0.000             | 0.000              | 0.000               | 0.000             | 0.000             |
| Covariates                              | Yes                 | Yes               | Yes                | Yes                 | Yes               | Yes               |
| Settlement/Year FE                      | Yes                 | Yes               | Yes                | Yes                 | Yes               | Yes               |

*Notes:* The table presents coefficients from an instrumental variables regression using the instrument described in the Materials and Methods section. These estimates are restricted to the sub-sample of households whose land allocation was less than their respective cohort average. All outcomes represent the number of days in the past 7 days the household employed the coping strategy. Each estimation includes the full set of household controls, weather controls, and fixed effects described in Table S1.

**Panel A (Dietary):** *Less Pref.* = Rely on less preferred/expensive foods; *Limit Size* = Limit portion size at mealtimes; *Red. Meals* = Reduce number of meals eaten in a day; *Restr. Adult* = Restrict adult consumption for children; *Red. Other* = Reduce consumption by others so workers can eat; *Skip Day* = Go an entire day without eating.

**Panel B (Livelihood):** *Immature* = Harvest immature crops; *Seed Stock* = Consume seed stock intended for next season; *Nat. Res.* = Sell natural resources (charcoal, firewood, etc.); *Sell Assets* = Sell household assets; *Barter* = Exchange goods for food; *Work 4 Food* = Exchange labor for food.

**Panel C (Social):** *Credit* = Purchase food on credit; *Borrow* = Borrow food from friends/relatives; *Wild Food* = Gather wild food or hunt; *Send Away* = Send household member to eat elsewhere; *Beg* = Beg for food; *School Food* = Send children to school specifically for meals.

Standard errors are clustered at the household level in parentheses.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .



Table S13: Leave-One-Out and Naive Results: Non-Imputed Observations

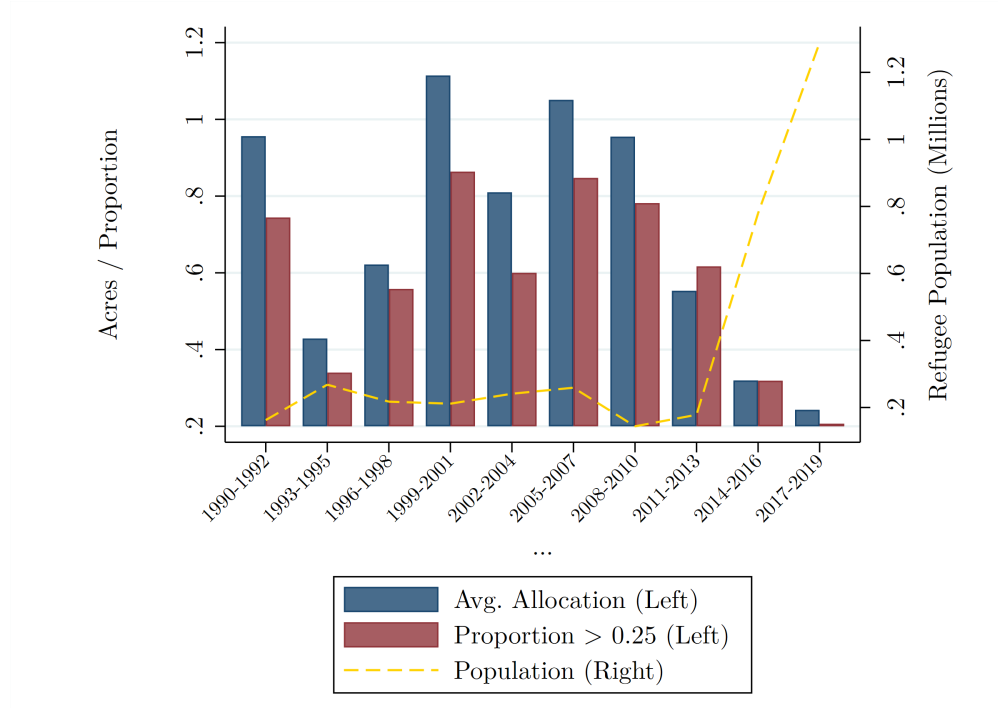
|                                    | (1)              | (2)                | (3)                | (4)                 | (5)               | (6)                 | (7)                 | (8)               | (9)               |
|------------------------------------|------------------|--------------------|--------------------|---------------------|-------------------|---------------------|---------------------|-------------------|-------------------|
|                                    | Food Security    | Any Harvest        | ln(Harvest > 0)    | Any Earnings        | ln(Earnings > 0)  | ln(Land Use)        | Any Livestock       | ln(Livestock > 0) | Asset Index       |
| <b>Panel A: IV (Leave-One-Out)</b> |                  |                    |                    |                     |                   |                     |                     |                   |                   |
| Gov't Land                         | 0.152<br>(0.154) | -0.027<br>(0.044)  | 0.604**<br>(0.260) | 0.154*<br>(0.079)   | 0.489<br>(0.381)  | 0.772***<br>(0.268) | 0.228***<br>(0.068) | 0.326*<br>(0.169) | 0.057<br>(0.135)  |
| Mean Dep. Var.                     | 0.07             | 0.87               | 4.38               | 0.28                | 2.60              | -1.28               | 0.48                | 1.12              | 0.01              |
| Obs.                               | 4407             | 4413               | 3638               | 4413                | 1044              | 3774                | 4413                | 2126              | 4413              |
| R-sq                               | 0.099            | 0.165              | 0.384              | 0.126               | 0.261             | 0.232               | 0.266               | 0.362             | 0.185             |
| WeakID (Wald F)                    | 49.00            | 48.93              | 47.44              | 48.93               | 31.91             | 48.00               | 48.93               | 41.90             | 48.93             |
| UnderID P-Val                      | 0.000            | 0.000              | 0.000              | 0.000               | 0.001             | 0.000               | 0.000               | 0.000             | 0.000             |
| <b>Panel B: Naive OLS</b>          |                  |                    |                    |                     |                   |                     |                     |                   |                   |
| Gov't Land                         | 0.054<br>(0.037) | 0.024**<br>(0.012) | 0.113*<br>(0.062)  | 0.050***<br>(0.018) | -0.084<br>(0.132) | 0.278***<br>(0.076) | 0.028*<br>(0.016)   | -0.041<br>(0.042) | -0.018<br>(0.035) |
| Mean Dep. Var.                     | 0.07             | 0.87               | 4.38               | 0.28                | 2.60              | -1.28               | 0.48                | 1.12              | 0.01              |
| Obs.                               | 4407             | 4413               | 3638               | 4413                | 1044              | 3774                | 4413                | 2126              | 4413              |
| R-sq                               | 0.101            | 0.169              | 0.399              | 0.135               | 0.279             | 0.244               | 0.292               | 0.388             | 0.186             |
| Covariates                         | Yes              | Yes                | Yes                | Yes                 | Yes               | Yes                 | Yes                 | Yes               | Yes               |
| Settlement/Year FE                 | Yes              | Yes                | Yes                | Yes                 | Yes               | Yes                 | Yes                 | Yes               | Yes               |

Notes: Panel A presents coefficients from an instrumental variables regression using the instrument described in the Materials and Methods section, while panel B presents the same estimates using OLS. The estimates exclude observations where missing control variables were replaced by group averages. Each column represents a separate regression with dependent variables as listed in the column headers. Both panels include the full set of household controls, weather controls, and fixed effects described in Table S1. Standard errors are clustered at the household level.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

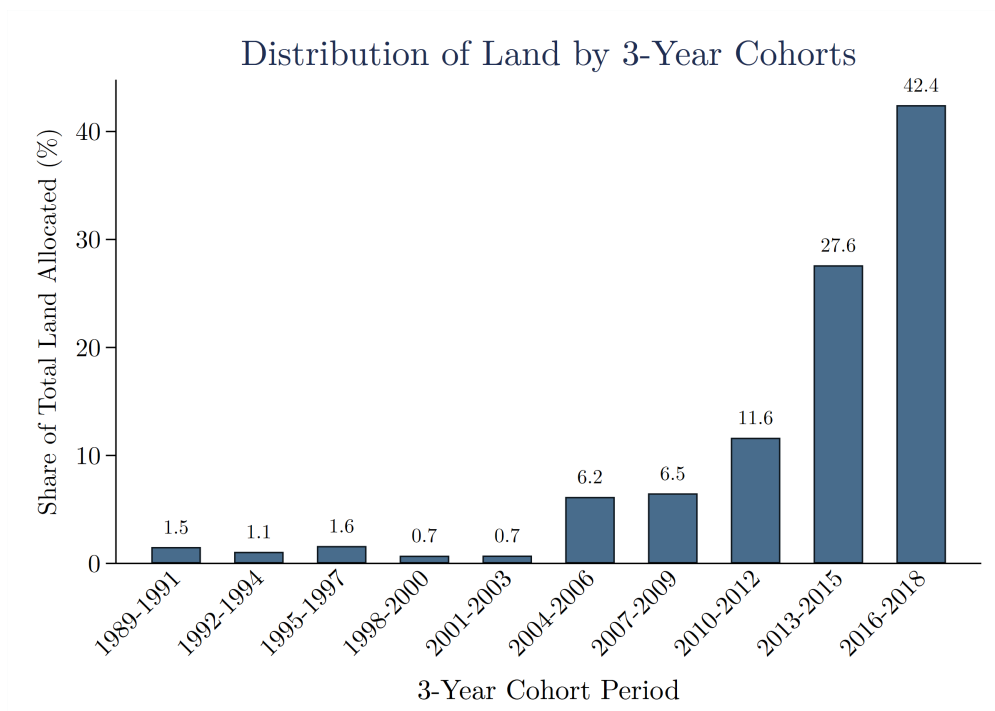
## C Supplemental figures

Figure S1: Land Allocation and Refugee Population by Arrival Cohort



*Notes:* Bars show the average land allocation in acres (blue) and the proportion of the population with an allocation over 0.25 acres (red) across arrival cohorts aggregated into 3-year groups (labelled on x-axis). The left y-axis provides the scale for the bars. The right y-axis provides the scale for total refugee arrivals, shown in the dashed line.

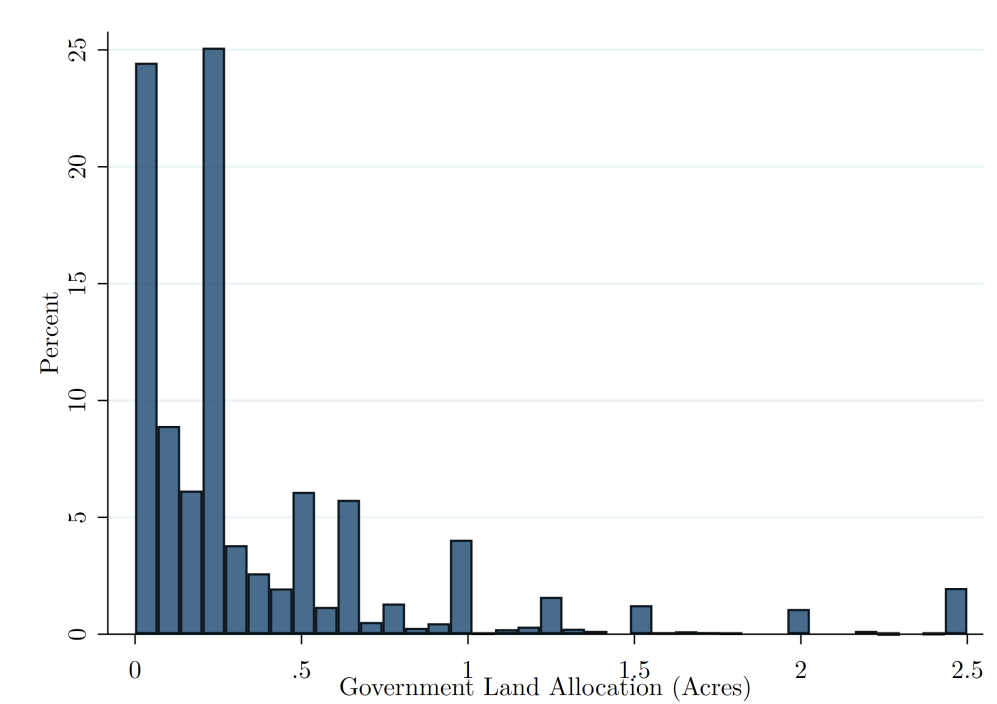
Figure S2: Proportion of Land Allocation Across Cohorts



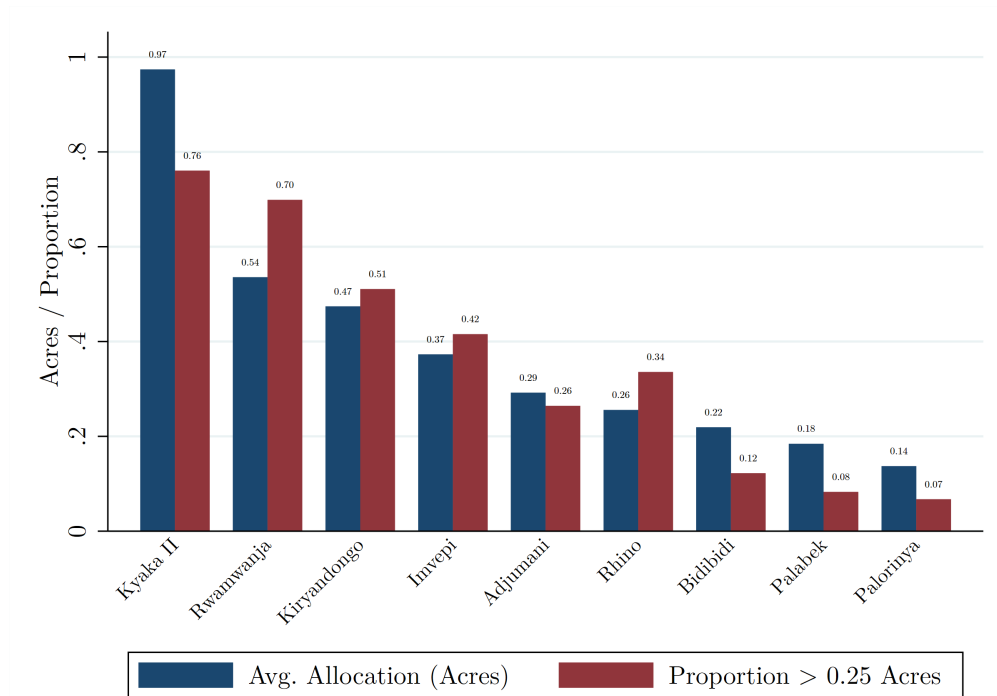
*Notes:* Bars show the proportion of total allocated land each 3-year cohort has been attributed (x-axis). The y-axis provides the total share of land allocated as a percentage.

Figure S3: Land Distribution Across Entire Sample and by Settlement

(a) Full Distribution of Land (Winsorized 95th Percentile)



(b) Across Settlements



**Notes:** **Panel (a)** shows the distribution of household land allocations in acres. The x-axis presents allocation sizes. The y-axis represents the percentage of households in the sample with said allocation. **Panel (b)** show the average land allocation in acres (blue) and the proportion of the population with an allocation over 0.25 acres (red) across settlements (labeled on x-axis). The left y-axis provides the scale for the bars.

## **D Structured interview instrument**

The following lists the questions asked, depending upon the identity of the interviewed person. Verbal consent was obtained before beginning the interview process.

### **For Settlement Officials and OPM (Office of the Prime Minister) Representatives**

- Roles and Responsibilities
  1. Could you describe your role and main responsibilities in regards to the process of settling refugees?
- Settlement Process
  1. Can you walk through the process a refugee goes through from when they arrive at the border to their settlement?
  2. Can you describe to me the process of how refugees end up in particular settlements, is there any element of choice for the refugees?
  3. How long do you expect refugees to stay in Uganda?
- Land Allocation and Use
  1. Can you describe how refugees are allocated land for agriculture? How is the size determined? Does this process differ across settlements?
  2. What are the biggest challenges regarding land allocation and management? e.g. availability, quality, disputes.
  3. Do you see the sale or trading of land frequently happening between households?
- Food Aid and Market
  1. Could you explain the current food assistance (ask about earlier if USAID/UN cuts significantly altered current program) program in the settlement? How is the amount and duration of assistance determined for a household?
  2. From your perspective, do you see food assistance altering the local food markets?
  3. What are the most common ways, apart from farming, do you see refugees earning an income?
  4. Do you believe that refugees are largely self-reliant? If not, in what ways do you believe these households are prevented from being so?
- Host Community and Integration
  1. How would you describe the relationship between refugee and host communities in the area?
  2. What kinds of public services can refugees access (e.g., schools, hospitals, agricultural subsidies, etc.)? What kinds of services are they not allowed to access that are available to Ugandan citizens? Are there any programs that Ugandan citizens cannot access that are available to refugees?

- Large Takeaways

1. What are the main challenges that you see with the refugee settlement program as it is currently operated?
2. If you had the resources, what is one thing you would change to improve the well-being of refugees here?

## **For Refugee Households**

- **Arrival and Settlement**

1. When did you arrive in Uganda and in this particular settlement?
2. Could you describe your path to this settlement? How was this settlement chosen?

- **Land and Farming**

1. When you arrived, were you given a plot of land for farming? Can you describe this process? How big was the plot of land, and has it changed in size or location since you've arrived?
2. How would you describe the quality of your land?
3. What crops did you decide to plant this season? What helped you decide?
4. Of the food you grow, how much do you keep for your household, and how much are you able to sell?

- **Food Security and Income**

1. In the past month, what have been the main sources of food for you?
2. In the past 6 months, have you received food assistance (either cash or in kind)? If in kind, do you consume all of the food, or do you sometimes sell it? Why or why not?
3. In the past 6 months, have you received assistance for agricultural production? Tell me about how you choose which crops to plant. Out of the food that you produce, what share of it do you sell? Where do you sell it?
4. Do you feel that you are able to satisfy your household's food requirements by the combination of food assistance and your own produced food? Or do you have other jobs to help buy food? What kind of jobs do you or others in your household have other than farming?
5. What do you find are the biggest challenges to getting enough food for your household?

- **Markets and Community**

1. When you sell your crops or buy food, where do you go? Can you describe the market?
2. Have you seen changes in the settlement when large groups of new refugees arrive? What kind of changes?
3. When new refugees arrive, does this change anything for your household? If so, in what way?
4. How would you describe your relationship with your Ugandan neighbors?